

Auditing LLM Evaluation with Imperfect Proxies

Targets, useful corrections, and failure modes

Siquan Wang

Technical report - retrospective empirical and methodological study

Evidence snapshot: [b4bdeab08481a25b529d81fa909f96aab10f368f](#). This report synthesizes existing experiments; it introduces no new estimator or theorem and is not presented as a peer-reviewed publication.

Abstract

An inexpensive LLM judgment can supplement a reference audit without being an accurate substitute for it. Whether that information helps depends on the outcome, target, proxy relationship and sampling design. We study these dependencies using established prediction-assisted mean estimators, public cached judgments and known-truth simulations. Empirical experiments hold evaluation groups fixed across nested audit budgets, while simulations separately measure point error and target-matched interval coverage. The results show modest, heterogeneous gains from adaptive correction, useful inverse signals, and substantial failures of fixed reliance on agreement. They also separate accuracy from uncertainty: in one near-boundary setting, tuned correction improves RMSE while a nominal 95% normal interval covers only 52.4% of repetitions. Applicable concentration bounds preserve their finite-sample interpretation at a substantial width cost. Population and random-pool targets can favor different coefficients, and neither tuning nor correct grouping repairs arbitrary distribution shift. The contribution is a reproducible comparison with explicit estimands, source provenance, paired evidence and retained failures. Historical, overlapping benchmarks and retrospective study development limit claims about new tasks, current judges or deployment savings.

1. Question and contribution

The central question is: **when does a frozen proxy improve estimation of a clearly defined reference mean under a limited audit, and what can be said separately about uncertainty?** A reference may be a recorded human preference or a benchmark's supplied correctness criterion. These meanings are not interchangeable. Likewise, estimating an aggregate mean is different from learning calibrated per-item probabilities or recovering latent truth.

Prediction-powered inference (PPI) corrects predictions using paired audit residuals [2]; efficient scalar tuning is developed in PPI++ [3]. The idea also has classical difference-estimation roots [1] and direct predecessors in automatic-metric correction for NLP [4]. We apply this established framework to LLM-evaluation audits, including agreement proxies whose relation to correctness may be weak or negative.

Three features make the comparison auditable. First, empirical methods share reference budgets and whole-group splits, with evaluation labels reserved for scoring. Second, simulation studies distinguish known population truth, random held-out means and deliberate assumption violations. Third, complete outcomes, input hashes, configurations and split identities remain available, including cases where correction loses. The analysis is retrospective: earlier findings motivated later choices. No design or result is described as preregistered.

The [evidence index](#) and [machine-readable crosswalk](#) connect quantitative claims to source artifacts, selectors, metric definitions and hashes. Full reports provide the surrounding grids; examples in this synthesis explain mechanisms rather than replacing those complete results.

2. One correction, different targets

Let Y be a bounded reference outcome and F a frozen proxy, both in $[0, 1]$. The audit contains n aligned outcome-proxy pairs. A separate pool contains N proxy observations whose reference outcomes are hidden from fitting. For coefficient λ , write:

$$\begin{aligned}\theta_{\text{hat}}(\lambda) &= \text{mean}(Y_L) \\ &\quad + \lambda * [\text{mean}(F_U) - \text{mean}(F_L)] \\ R(\lambda) &= Y - \lambda * F\end{aligned}$$

Zero power gives the reference-audit mean; power one gives the usual mean correction. Fixed-power unbiasedness requires an unbiased audit outcome mean and equal expected proxy means across pools. Independent draws from the same relevant population with a frozen predictor supply a sufficient setting. More prediction observations do not remove audit-selection bias or a changed judge mechanism.

For a population target $\theta = E[Y]$, independent iid pools give:

$$\begin{aligned}\text{Var}(\theta_{\text{hat}} - \theta) &= \text{Var}(R)/n + \lambda^2 * \text{Var}(F)/N \\ \lambda_{\text{population}} &= [N/(n+N)] * \text{Cov}(Y, F)/\text{Var}(F)\end{aligned}$$

The second expression is an unrestricted oracle when proxy variance is positive. The implementation estimates a separate variance for each pool and projects the resulting minimizer onto $[0, 1]$, or explicitly onto $[-1, 1]$ for signed numeric inference. Exact constant proxies receive zero power. The fitted variance cannot exceed its zero-power endpoint, because zero is feasible. This optimization fact does not guarantee lower realized loss, smaller actual finite-sample variance or valid coverage.

For the random target $\text{mean}(Y_U)$, the estimation error instead equals the difference of two residual means. The target shares proxy information with the estimate, so their covariance matters:

$$\begin{aligned}\theta_{\text{hat}} - \text{mean}(Y_U) &= \text{mean}(R_L) - \text{mean}(R_U) \\ \text{Var}(\theta_{\text{hat}} - \text{mean}(Y_U)) &= (1/n + 1/N) * \text{Var}(R) \\ \lambda_{\text{pool}} &= \text{Cov}(Y, F)/\text{Var}(F)\end{aligned}$$

These are fixed-coefficient iid identities. Fitted coefficients remain plug-in procedures with asymptotic interpretations. The associated pool interval is a marginal prediction interval over draws of both pools, not conditional coverage for every frozen pool. A perfect proxy illustrates the distinction: power one recovers its pool's outcome mean exactly, whereas population tuning combines information from both samples.

Fixed finite populations require their sampling design to supply the randomness [5]. Under disjoint simple random partitions and fixed power, negative cross-pool covariance cancels the familiar finite-population corrections in the residual-mean difference. That identity does not make adaptive tuning exact, and substituting row counts does not establish a theorem for unequal prompt clusters. The empirical audit-residual candidate therefore returns a point without a pool interval.

Grouped population diagnostics use centered cluster totals in a one-way sandwich, retaining observation weights. They require sufficiently many independent groups and suitable regularity, not merely valid group IDs. The [Methods](#) document gives equations, numerical conventions and distinctions from the authors' `ppi-python==0.2.3` implementation. The pinned comparison checks fixed-weight algebra and expected finite-sample convention differences; it is not independent evidence of coverage.

3. Evidence and experimental design

The empirical and simulation components answer different questions. Their metrics are kept separate throughout this report.

Evidence	Outcome and scored target	Sampling unit	Important overlap
MT-Bench	Recorded preference; held-out mean	Complete question	Same data for reference sensitivity
LLMBar	Judge reference-correctness; held-out mean	Exact instruction	Signed and pool follow-ups reuse it
RewardBench	Judge reference-correctness; held-out mean	Exact prompt	Includes 419 LLMBar comparisons
Population simulation	Defined outcome; analytic $E[Y]$	Specified iid rows or clusters	Related generator families
Pool simulation	Defined outcome; random mean(Y_U)	Independent iid pools	Both targets use identical draws

MT-Bench provides 1,814 retained comparisons and 3,354 unique human votes across 80 questions. All 15 eligible model pairs are retained. A plurality outcome scores an A win, B win or tie as 1, 0 or 0.5. A separate analysis uses each comparison's mean recorded vote score. Averaging comparisons equally does not pool all votes into one vote-weighted target. The frozen categorical GPT-4 score supplies the proxy in both analyses.

LLMBar supplies 419 instruction-following comparisons and 418 exact instruction groups for three historical judges. Its proxy records whether both presentation orders produce valid, identical canonical choices. Invalid outputs remain in denominators: invalid original-order choices are incorrect, and invalidity in either order makes agreement zero. The upstream cache already canonicalizes reversed choices; extraction does not reverse them a second time.

RewardBench uses complete binary caches for GPT-4o 2024-08-06 and GPT-4o mini 2024-07-18. For two judges choosing between the same two answers, equality of their correctness bits reconstructs canonical choice agreement. Reversing the common reference complements both bits but leaves equality unchanged. An individual auxiliary correctness bit would retain gold information and is not used as a predictor. A third inspected Gemini cache has nonbinary fallback scores and is excluded as a whole under this binary contract; no comparisons are removed from the selected panel.

The RewardBench primary component excludes LLMBar: 2,566 comparisons and 2,315 prompt groups. The full mixture and LLMBar component are sensitivity analyses, not independent replications. Complete content and reference checks establish the 419-record overlap. Remaining sources include MT-Bench-derived material, so excluding LLMBar does not certify absence of every shared benchmark lineage. RewardBench row means also differ from its official weighted leaderboard score.

Each empirical study uses 30 deterministic splits. Evaluation contains $\text{floor}(0.25 \cdot G)$ complete groups; nested audits contain $\text{floor}(b \cdot G)$ groups for b equal to 0.20, 0.40 or 0.60, where G is the total eligible group count. Fractions are not percentages of the remaining bank. Larger audits leave evaluation fixed. Repeated questions or exact prompts stay together, including cross-subset repeats. Unused rows do not enter fitting. Exact grouping does not establish independence of semantic duplicates, related tasks or shared annotators.

Errors are scored against the realized held-out reference mean. Population interval widths are diagnostics, not coverage tests for that target. Empirical splits overlap, so their averages receive no iid Monte Carlo standard error. Synthetic replications are independent within each specified scenario and retain paired method contrasts and Monte Carlo uncertainty. The historical complementary-pool MT-Bench study, whose evaluation size shrinks with audit budget, remains separate.

4. Findings

4.1 Target and reference definitions change the comparison

In the positive-proxy simulation with prevalence 0.5, 200 audited rows and 100 prediction rows, replacing population tuning with pool tuning changes population-target RMSE from 0.0309 to 0.0462. Against the pool target, RMSE changes from 0.0472 to 0.0323. These are two within-target comparisons with opposite directions, not evidence that one target is intrinsically easier or more appropriate. Paired squared-error differences preserve the shared draws and appear with Monte Carlo errors in the [estimand report](#). Evidence E07 records the exact cell and target-specific columns.

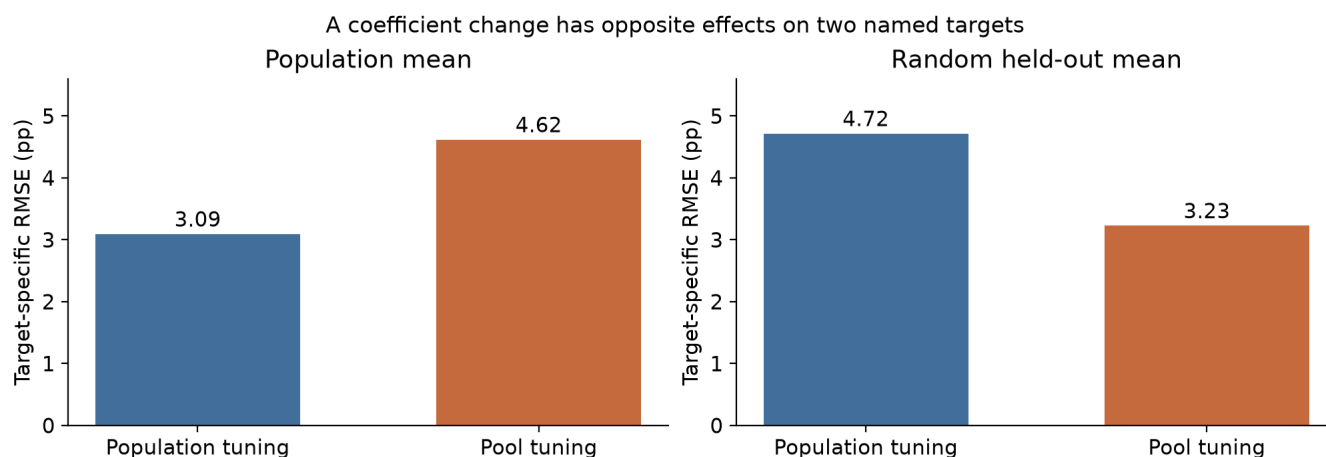


Figure 1. Existing simulation results for the positive proxy, prevalence 0.5, $n=200$ and $N=100$. Compare the two rules within each target's RMSE panel; the opposite directions illustrate the target distinction. This selected case does not summarize the complete grid or describe empirical coverage.

Matching the oracle criterion is insufficient after fitting. For the positive proxy at prevalence 0.95 with 20 audit and 10 prediction rows, pool tuning worsens pool RMSE from 0.0728 to 0.0762; its nominal 95% marginal prediction interval covers in 77.4% of repetitions. The complete 36-setting study retains these losses and every degenerate interval.

The outcome definition also matters. On MT-Bench, plurality and mean-vote scores differ on 305 comparisons; 854 comparisons have only one vote. Tuned PPI retains a modest aggregate MAE advantage over the reference audit at all budgets under both definitions, but the direction of the pair-level comparison changes in 5 of 45 matched pair/budget cases. Each method is scored against its own named reference. A lower error under one definition does not establish better truth recovery (E02). Vote counts remain necessary because the same mean can represent unanimous ties or polarized choices.

4.2 Useful correction is target-specific and often modest

In the fixed-target MT-Bench analysis, tuned PPI improves aggregate MAE over the reference-only audit by approximately 0.362, 0.238 and 0.208 percentage points at the three budgets. It does not improve every pair: the lower-MAE counts are 10/15, 9/15 and 11/15. Fixed coefficient-one PPI has higher aggregate MAE than the audit at each budget. These are descriptive results on the selected pair/split cases, not a population probability of improvement or an annotation-saving guarantee (E01).

Agreement has even more direct limitations. On LLMBar Adversarial, ChatGPT's order agreement is 64.3% while original-order reference accuracy is 28.2%. Of 205 consistent comparisons, 176 consistently choose the reference-incorrect answer (E08). Consistency can therefore be confidently wrong. For ChatGPT and LLaMA2 on this component, positive-only tuning selects zero on every split and falls back to the audit; fixed-power correction worsens MAE at every budget. The benchmark's ChatGPT-influenced adversarial

construction limits any general ranking of judge families.

RewardBench exposes the target dependence of one shared proxy. In the primary NonLLMBar component, agreement is 89.95%, while reference accuracy is 90.80% for GPT-4o and 86.67% for mini. Both judges agree on the wrong answer in 160 comparisons (E09). Positive and signed population tuning have identical results here and improve MAE in all six judge/budget cells, whereas fixed coefficient one worsens five. The improvements range from only about 0.011 to 0.252 percentage points; a favorable sign alone does not establish practical importance (E10).

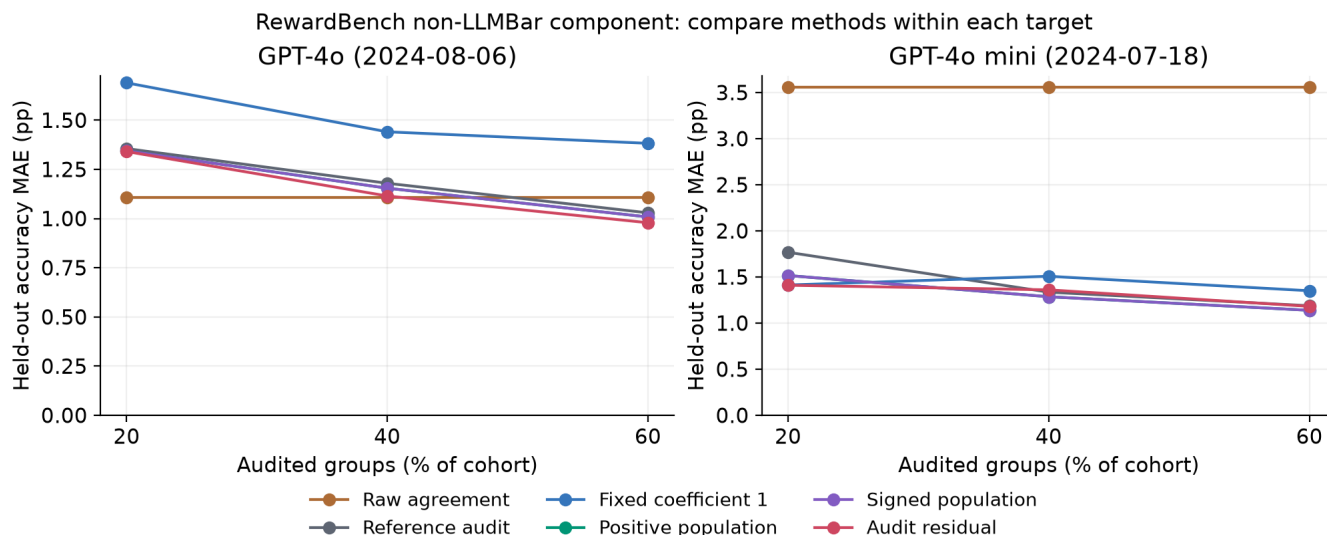


Figure 2. Existing NonLLMBar results for both designated judges, all six rules and all three audit budgets. Compare methods within the same target; panel scales may differ. Positive and signed population curves coincide in these primary data. Full-mixture and LLMBar sensitivities remain in the complete report. Reused splits are not independent replications.

Counterexamples remain important. For GPT-4o at the lowest primary budget, raw agreement has MAE 1.107 percentage points, versus 1.356 for the audit and 1.345 for population tuning. Correction does not uniformly beat the raw surrogate. At the 40% budget, audit-residual tuning improves GPT-4o MAE from 1.179 to 1.115 points but worsens RMSE from 1.421 to 1.432. MAE and RMSE summarize different losses; presenting only the favorable metric would overstate the evidence (E10).

4.3 Signed flexibility uses inverse signal and can fit noise

Negative coefficients are already allowed by the PPI++ mean framework [3]. Applying negative weight to F is algebraically equivalent to positive weight on $1-F$. The signed option expands the declared search range to $[-1, 1]$; it does not turn agreement into truth or erase the uncertainty of choosing a coefficient from the audit.

The signed study makes the tradeoff concrete. With an inverse proxy, prevalence 0.5 and 20 audit observations, signed tuning reduces population RMSE from 0.1098 to 0.0683 relative to positive-only tuning (E05). With an uninformative proxy at the same prevalence and audit size, RMSE increases from 0.1108 to 0.1125 (E06). All methods share draws, and paired squared-loss Monte Carlo errors are available in the [complete signed study](#).

On unchanged LLMBar splits, signed tuning improves MAE over positive-only tuning in 11 of 27 cells, ties in 12 and worsens four. This follow-up was motivated by the earlier inverse association and reuses the corpus; it is exploratory evidence, not independent validation. Same-audit coefficient bias and finite-sample costs have prior treatments in NLP evaluation [4] and PPI analysis [6]. The results illustrate those concerns under this bounded, constrained implementation rather than discovering them anew.

4.4 Better point error does not validate uncertainty

The finite-sample study separates point estimation from interval coverage. For a strong proxy, prevalence 0.95 and 20 audit observations, tuned normal PPI has RMSE 0.0413 versus 0.0496 for the audit-only normal procedure. Nevertheless, its nominal 95% interval covers in only 52.4% of 1,000 repetitions. The pointwise exact Monte Carlo interval is approximately [49.3%,55.5%], and 353 intervals have zero width. A sample with no observed outcome variation has not established population certainty (E03).

Established bounded-data inequalities provide a different guarantee under their assumptions. The implementation uses fixed powers or a prespecified finite grid, allocating error simultaneously across candidate residual bounds and a shared prediction-mean bound. It applies [Hoeffding's inequality](#) and [Maurer-Pontil empirical Bernstein bounds](#), with constants and allocation derived in the [finite-sample protocol](#). PPI itself already contains nonasymptotic mean-inference constructions [2]. These applications are not new concentration results.

In the same small-audit setting, audit-only Hoeffding covers in all 1,000 draws, with an exact coverage Monte Carlo interval approximately [99.63%,100%]. Mean untruncated width is 0.607, versus 0.093 for tuned normal. Grid empirical Bernstein width is 1.924 and contains the entire $[0, 1]$ parameter range in 64.7% of repetitions. Conservative coverage can be uninformative. Across the twelve valid iid settings, grid Bernstein is narrower than audit-only Hoeffding in only one (E04).

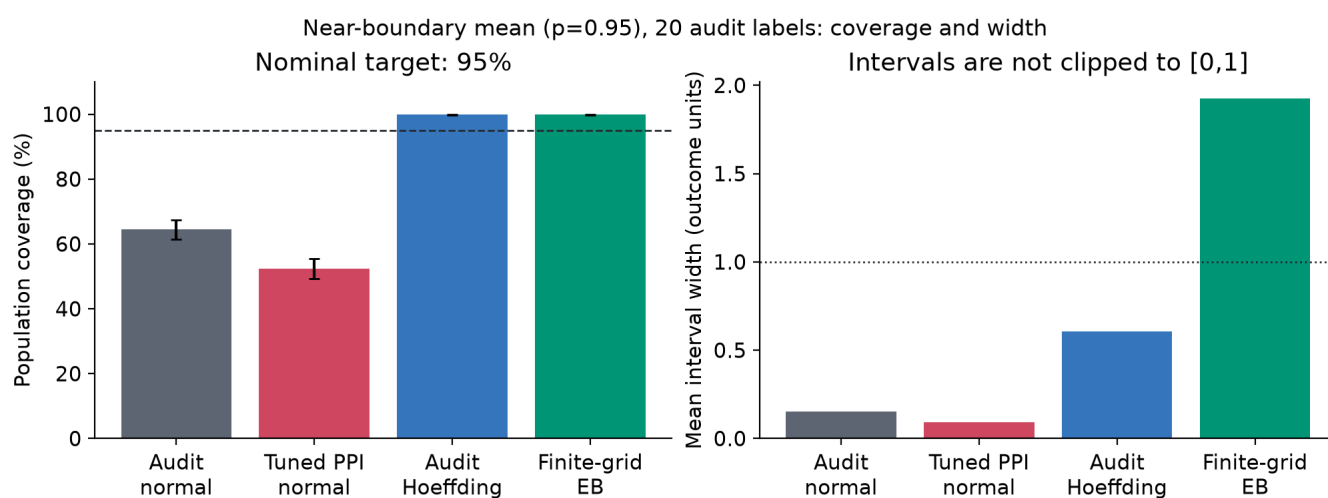


Figure 3. Existing iid strong-proxy results at prevalence 0.95, $n=20$ and $N=200$: audit-only normal, tuned normal, audit-only Hoeffding and grid empirical Bernstein. Observed coverage comes from 1,000 repetitions; error bars are pointwise 95% exact-binomial Monte Carlo intervals. Widths remain untruncated. The full report retains all eight methods, twelve iid settings and separate deliberate assumption violations.

A run without misses does not prove universal coverage. The mathematical bounds require independent iid pools, bounded scores, fixed sizes and a frozen predictor. They do not validate continuous same-audit tuning, dependent rows or transport to another population. Perfect observed predictions cannot replace the known residual range with the sample range. Intervals remain untruncated so their width cost is visible.

4.5 Grouping and transport remain separate requirements

The dependence ablation analyzes identical generated data with correct question grouping and deliberately naive row independence. With eight independent questions and four exact repeated turns per question, tuned coverage is 89.0% under grouping and 59.6% under the naive analysis. Grouping removes a major source of overconfidence, but eight clusters still do not make a normal approximation reliable. This is a controlled example, not a universal required number of clusters.

Shift creates a different problem. In the declared prevalence-shift scenario, tuned correction has bias about -0.1975 and zero target coverage; under a changed conditional judge-error mechanism, bias is about -0.0545 and coverage 23.9%. The fitted variance criterion cannot identify missing transport assumptions. These are failures of the basic correction under intentional violations, not refutations of PPI's separate shift procedures [2] or of methods designed for invariant conditional misclassification mechanisms [8]. The [stress report](#) and [initial simulation report](#) retain exact generators and all outcomes; E11 traces the dependence and shift examples.

5. Related work and claim boundaries

Classical generalized difference and regression estimation use auxiliary information for finite-population estimation [1]. Chaganty, Musmann and Liang [4] apply control variates to automatic NLP metrics and discuss annotation variation and bias from fitting coefficients on the same data. This lineage rules out claims that reference-audit correction or its plug-in bias originates here.

PPI [2] supplies the modern prediction-assisted inference framework, including mean, finite-population and nonasymptotic settings. PPI++ [3] develops efficient tuning, including real-valued mean coefficients. This repository's per-pool variance convention and optional cluster moments are explicit adaptations, not direct proofs of general cluster validity. Li and Ding [5] clarify finite-population sampling distributions; their results do not automatically supply conditional intervals for an arbitrary frozen or unequal-cluster panel.

Mani and colleagues [6] analyze finite-sample costs of tuned PPI, including same-sample bias and optimistic variance estimates. Their distributional regimes are not transferred mechanically to coefficient-constrained, bounded-outcome or grouped variants here. R-AutoEval+ [7] studies adaptive autoevaluation with sequential reliability and selection guarantees. Our fixed-budget means do not claim anytime validity or superior sample efficiency. Lee and colleagues [8] study misclassification calibration and uncertainty, with transfer conditions tied to the conditional error mechanism. A fair comparison would first align outcomes, decisions, sampling, stopping and costs; this report does not manufacture one from unrelated interval widths.

6. Limitations and interpretation

The principal limitation is external validity. Selected 2023/2024 caches, curated source mixtures and overlapping benchmark lineages do not establish performance for contemporary endpoints or a future evaluation workload. The two RewardBench judges share a model family. LLMBAR's construction partly used ChatGPT evaluators. Multiple reports on these records are not multiple independent confirmations, and a larger synthetic grid is not a random sample of deployments.

Reference uncertainty also remains unresolved. Recorded votes depend on the observed annotators and uneven annotation counts. The derived MT-Bench snapshot lacks per-vote annotator identities, preventing an assessment of shared annotator effects. Benchmark correctness combines several source criteria and is not latent truth. Changing a reference changes the estimand rather than correcting it automatically.

The empirical target is a realized held-out mean from an unequal-group finite corpus. We report point errors without claiming population or pool coverage there. Exact prompt groups address visible repeats, not every dependence structure. Small empirical gains and conflicting MAE/RMSE directions require restrained interpretation. No universal minimum audit size, optimal deployment budget or reliable stopping policy follows.

Finally, costs are logical inputs to a cached-data analysis. One reference reveal can score every judge on that comparison, so labels are shared. Agreement may require two cached judgments where an audit-only method needs one designated judgment. Evaluation references used to score the experiment are distinct from inference

inputs. No new annotations, tokens, API calls, time savings or dollar savings were measured.

The supported implication is procedural: name the reference and target, preserve dependence units, estimate proxy reliance using permitted audit information, and assess uncertainty separately from point error. It is not a validated rule that every audit should use a tuned correction.

7. Reproduction and evidence preservation

The evidence crosswalk records exact selectors, units and directions for the numbered findings. Each underlying report retains trials, summaries, split manifests, configurations and source/artifact hashes. No failed, zero-width or out-of-domain estimate is removed. Synthetic coverage uses known targets; the estimand study additionally documents a machine-precision membership margin that changes no point, variance or interval endpoint.

From the pinned repository snapshot, a representative offline reproduction is:

```
pip install -e "[dev,report]"
python examples/estimand_study.py --output reports/reproduced/estimand --plot
python examples/rewardbench_audit.py --output reports/reproduced/rewardbench-audit --plot
```

Other complete commands are retained in the [research](#), [reference-sensitivity](#), [signed](#) and [finite-sample](#) reports. Smaller smoke runs explicitly record their reduced repetition or seed counts.

Committed derived inputs omit raw prompts, answer text and private data. Source notices retain distinct dataset and cache terms, including the RewardBench results collection's unspecified named license. Pinning cached bytes reproduces this reanalysis, not an undocumented historical API invocation. Historical reports remain separate evidence artifacts; this synthesis does not silently replace their targets, sampling designs or results with newer method variants.

References

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